

智能科研方法

AI for Research: Methods and Practice

AI 可以参与科研，研究责任由人承担。

2 学分 | 32 学时 | 研究生专业选修课

饮水思源 · 爱国荣校



这段 AI 输出，在研究里算什么？

最近一次科研相关的 AI 输出

线索

草稿

证据

结论

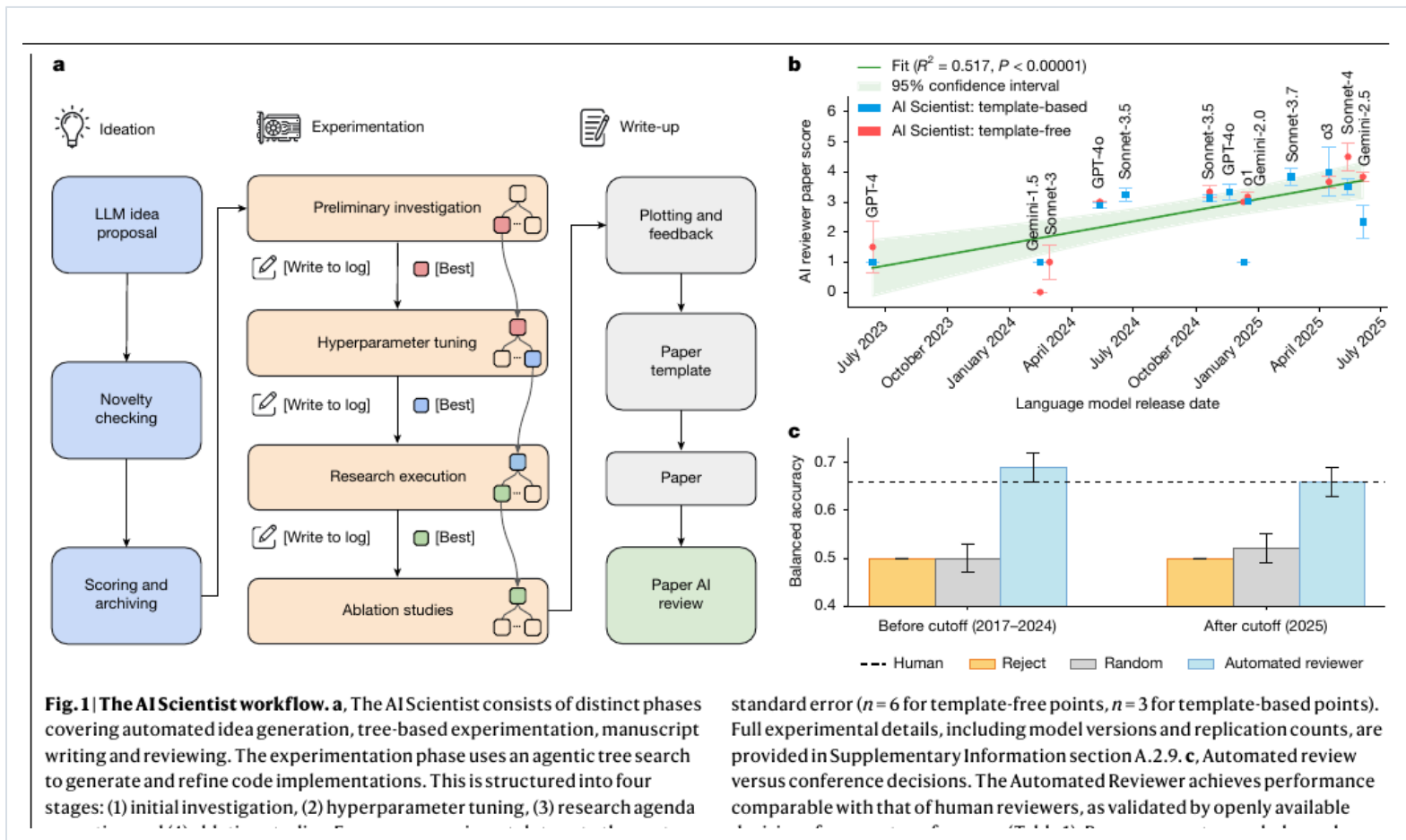
你把它升级到下一种角色之前，做过什么核验？

角色不是由语言流畅度决定，而是由核验过程决定。





AI Scientist 已串起完整研究流程



Lu et al. · Nature 651 · 2026 · Fig. 1

01 实验使用 tree search
而不是一次生成

02 中间结果写入 log
供后续阶段使用

03 评审也由模型组件执行
评价本身需要被检查





完整论文产物仍不足以证明研究成立

Towards end-to-end automation of AI research

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The automation of science is a long-standing ambition in artificial intelligence (AI) research^{1,2}. Although the community has made substantial progress in automating individual components of the scientific process, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach. Here we present a pipeline for automating the entire scientific process end to end. We present The AI Scientist, which creates research ideas, writes code, runs experiments, plots and analyses data, writes the entire scientific manuscript, and performs its own peer review. Its ideas, execution and presentation are of sufficient quality that the manuscript generated by this AI system passed the first round of peer review for a workshop of a top-tier machine learning conference. The workshop had an acceptance rate of 70%. Our system leverages modern foundation models^{3–5} within a complex agentic system. We evaluate The AI Scientist in two settings: a focused mode using human-provided code templates as an initial scaffold for conducting research on a specific topic and a template-free, open-ended mode that leverages agentic search for wider scientific exploration^{6,7}. Both settings produce diverse ideas and automatically test, report on and evaluate them. This achievement demonstrates the growing capacity of AI for making scientific contributions and signifies a potential paradigm shift in how research is conducted. As with any impactful new technology, there could be important risks, including taxing overwhelmed review systems and adding noise to the scientific literature. However, if developed responsibly, such autonomous systems could greatly accelerate scientific discovery.

AI has long been used to aid scientific discovery, an ambition with deep roots in the history of the field^{1,8–11}. Before the rise of large language models (LLMs), AI was limited to helping with specific, narrow tasks, such as discovering chemical structures², finding mathematical proofs⁵, discovering new materials^{12–14} and predicting the three-dimensional shape of proteins^{15,16}. Other systems focused on analysing pre-collected datasets to find new insights^{10,17,18}. However, with the recent advent of powerful and general foundation models, the role of AI has expanded to include assisting with a wider array of research activities. For example, LLMs now help with generating new hypotheses^{19–23}, writing literature reviews^{24,25} and coding experiments^{26–28}. Despite these advances in automating individual components, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach until now.

This paper introduces The AI Scientist, a pipeline that achieves the vision of full end-to-end automation of the scientific process. The AI Scientist uses existing foundation models to perform ideation, literature search, experiment planning and implementation, result analysis,

A central challenge in developing such a system is automatically evaluating the quality of its scientific output at scale. To address this, we created an automated reviewer and first evaluated its performance against real, human-generated papers. The Automated Reviewer can accurately predict conference acceptance decisions, performing on par with human reviewers (Supplementary Information section A.3). We then used The Automated Reviewer to compare various configurations of The AI Scientist by assessing how performance changes with the scale of the test-time compute and the quality of the underlying foundation model. We find that The AI Scientist performs better with more compute resources (Fig. 3c). Furthermore, The Automated Reviewer shows that improvements to the base models significantly improve the quality of the generated papers, a finding that strongly implies that future versions of our system will be substantially more capable, as models continue to improve (Fig. 1b).

To assess The AI Scientist in the same setting in which human-authored papers are evaluated, we conducted an experiment where we submitted generated papers to a workshop at the International Conference

3 → 1

系统稿件提交

达到 workshop 接收线

70%

ICBINB workshop 接受率

32%

ICLR 2025 主会议接受率

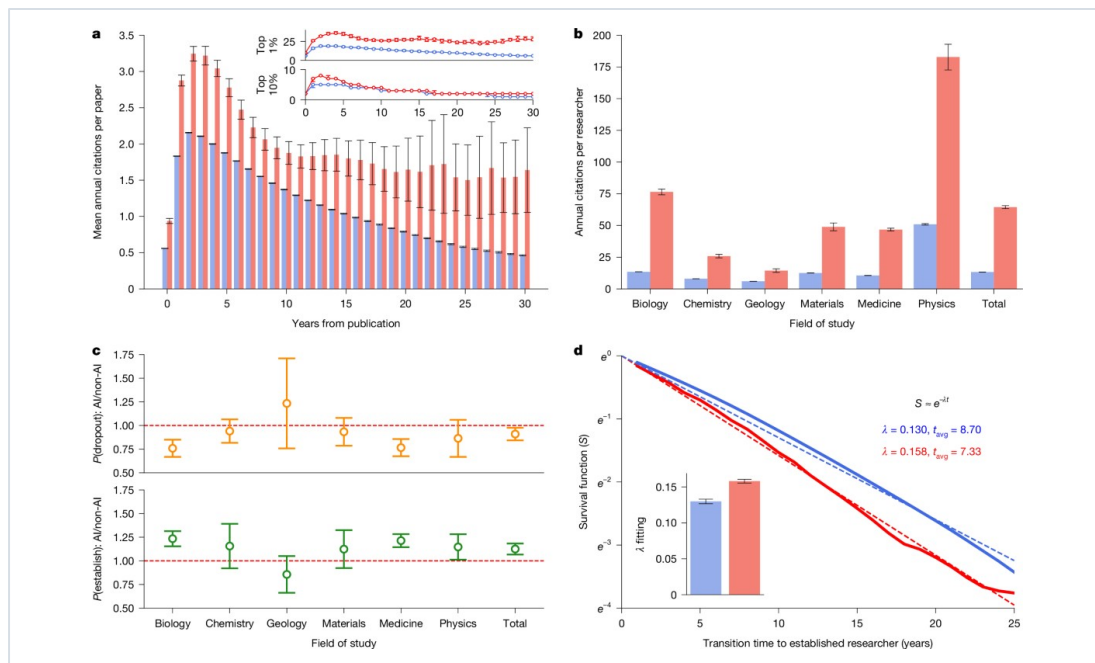
作者团队的内部判断：
3 篇均未达到 ICLR 主会议标准。





更高产，也可能让研究更集中

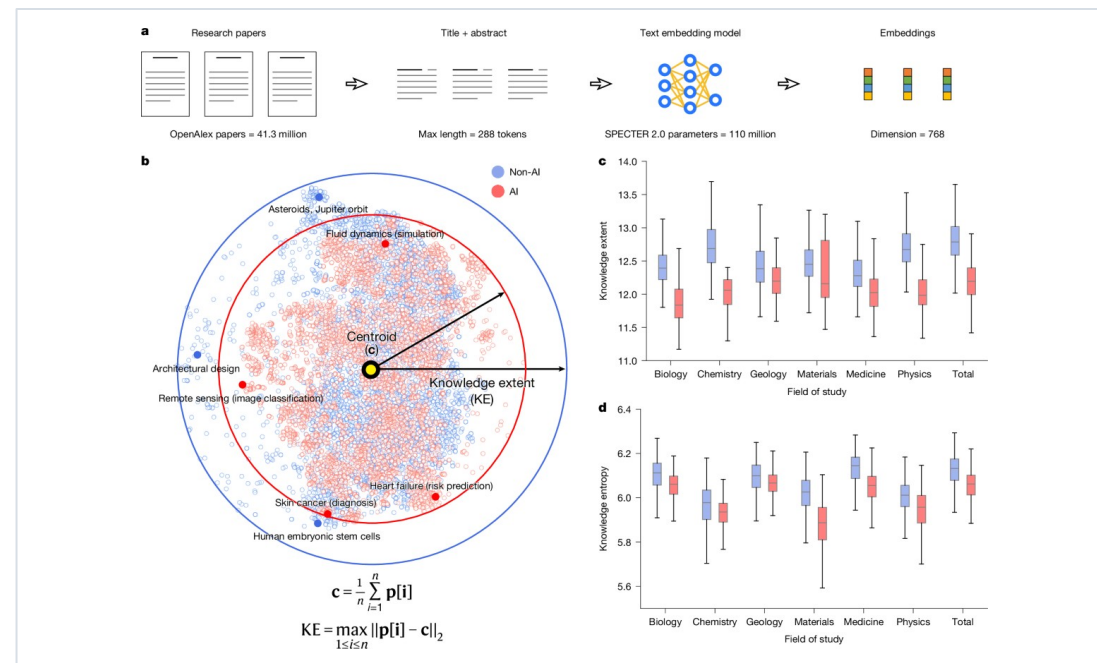
个人层面



3.02×

4.84×

集体层面



-4.63%

-22%

基于 4130 万篇论文的观察性关联 | 数字不构成因果证明





研究质量来自三种控制

AI 扩展候选，研究者控制升级条件。



可审计的研究闭环

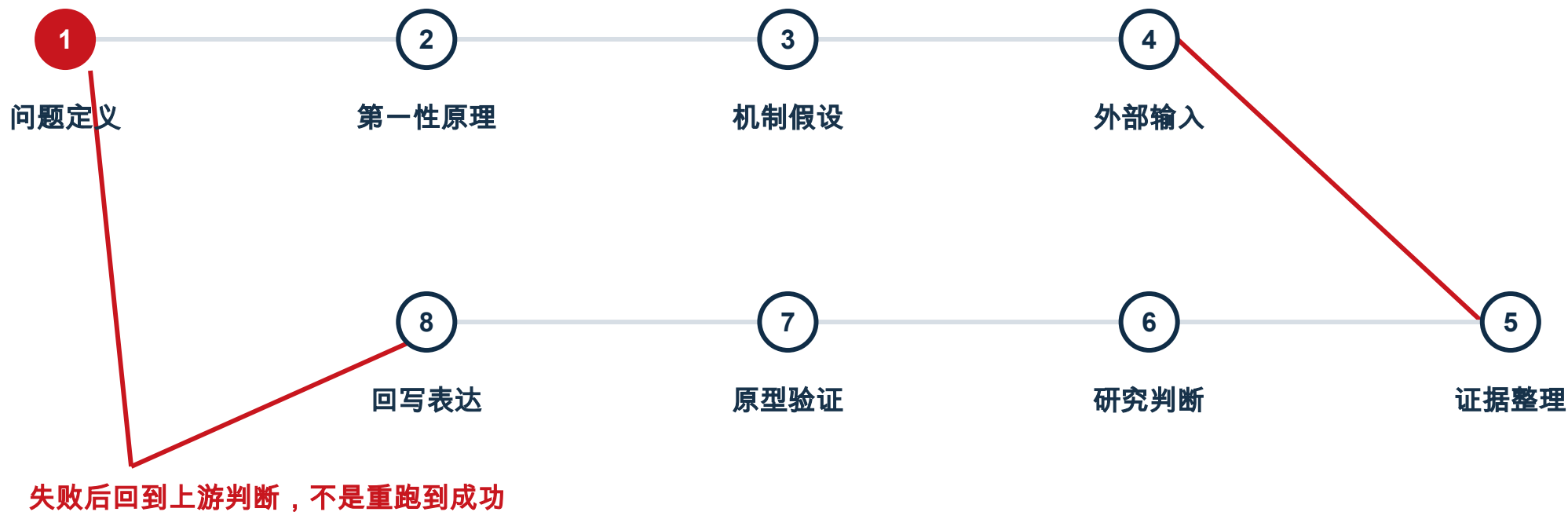
可核验回答“依据在哪里” | 可复现回答“能否重做” | 可追溯回答“谁做了什么判断”





用一条 claim 走完整个研究闭环

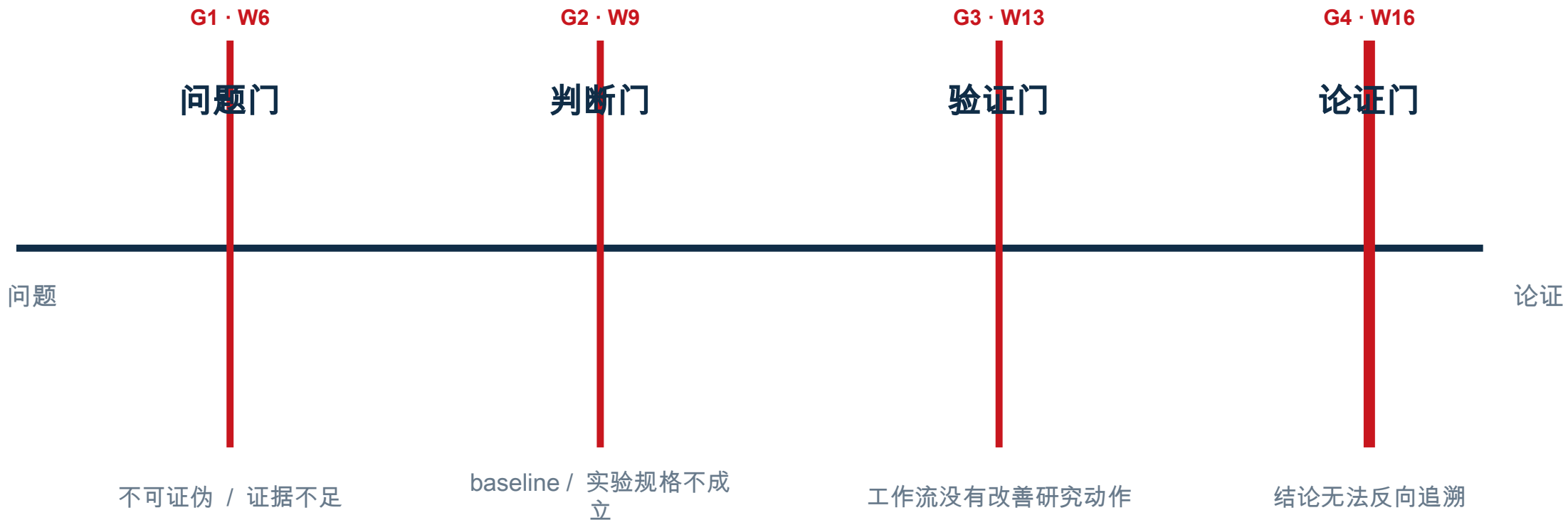
示例 claim：方法 A 在条件 C 下优于 baseline B。





四个研究门分别拦住四类错误

门的作用：在错误进入下一阶段前，要求补证据或退回修改。





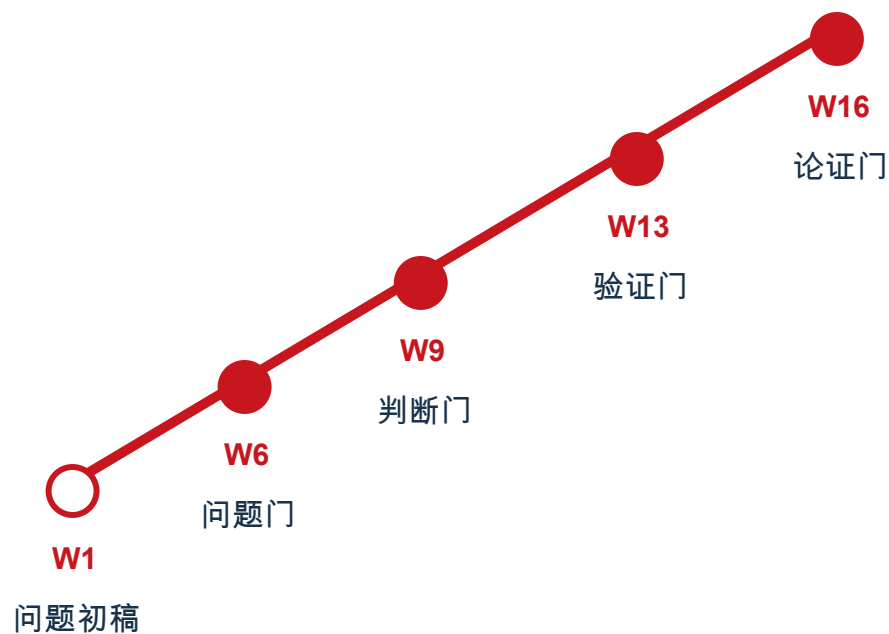
16 周只维护一个个人项目

starter-template.md · 最小文件结构

```
project/
├── problem-definition.md
├── reading-cards.md
├── artifact-tracking.md
├── ai-usage-log.md
├── agent-permissions.md
├── ethics-and-compliance.md
└── notes/
```

第 1 周允许它不完整；后续课堂持续回写同一项目。

一个项目，持续生长





五类工件最终汇成一条论证链



结论必须能反向追溯到实验、证据和当时的研究判断。



课程沿四个研究门推进同一个项目

前半段刻意放慢：问题和证据没站稳，自动化只会更快地产生不可用结果。



同一个个人项目持续回写



评分落在四类可检查证据上

文献与问题定位

证据地图 / 可证伪命题



25%

实验设计与可复现性

baseline / 实验规格



30%

原型与 Agent Workflow

评价报告 / 失败分析



25%

表达、伦理与复盘

可追溯论证 / AI 披露



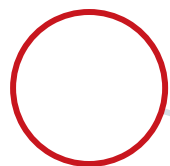
20%

评分证据来自四个研究门；模型价格、Agent 数量和 Skills 数量不进入评分。

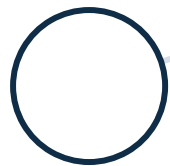




个人负责判断，同伴负责质询



项目 A

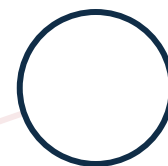


项目 C

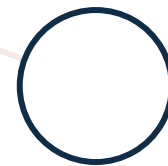
同伴质询记录

- 哪项证据最薄弱？
- 什么结果会推翻当前假设？
- 下一步实验能改变哪个判断？

采纳 / 拒绝 / 暂缓
均由项目作者说明理由



项目 B



项目 D

独立问题 · 独立判断 · 独立提交





今天离开前，留下三个可检查工件

problem-definition.md

研究问题
我们研究在 _____ 条件下，
_____ 方法是否能改善
_____。

范围： _____
非例： _____

01

八阶段链路草图

阶段	当前想法
1 问题定义	_____
2 第一性原理	_____
3 机制假设	_____
...	
8 回写表达	_____

02

ai-usage-log.md

工具 / 模型： _____
用途： _____
Context： _____
输出用途： _____
人工核验： _____
问题或误导： _____

03

不求完整，只求下一步可以被检查。

